

HINTS & SOLUTION

Q.1 (C)

Use $P_A = C_A RT$ and $C_A = \frac{n_A}{V}$

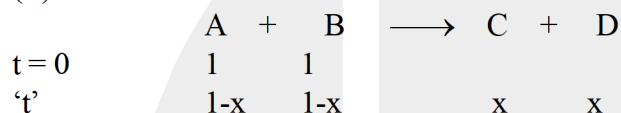
Q.2 (C)

Q.3 (B)

Overall order of photochemical reactions is not always zero.

Q.4 (A)

Q.5 (B)



rate = $k[A]^{1/2}[B]^{1/2}$

$$-\frac{d[A]}{dt} = k(1-x)^{1/2}(1-x)^{1/2} \Rightarrow -\frac{d(1-x)}{dt} = k(1-x) \Rightarrow \frac{dx}{dt} = k(1-x)$$

$$\Rightarrow \int_0^x \frac{dx}{1-x} = k \int_0^t dt \Rightarrow [-\ln(1-x)]_0^x = kt \Rightarrow -\ln(1-x) = kt$$

$$\Rightarrow -\ln(1-x) = kt \Rightarrow t = \frac{1}{k} \ln\left(\frac{1}{1-x}\right)$$

When $[A] = 0.25$

$\therefore x = 0.75$

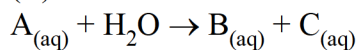
$$\therefore t = \frac{1}{2.31 \times 10^{-3}} \ln \frac{1}{0.25} = \frac{10^3}{2.31} \ln 4 = \frac{2 \times 10^3 \times 0.693}{2.31} = \frac{2 \times 10^3 \times 0.693}{2.31} = 600 \text{ Sec.]}$$

Q.6 (C)

Q.7 (C)

Q.8 (D)

Q.9 (C)



$$\text{Rate} = K[A][H_2O]$$

Reaction is pseudo first order reaction w.r.t. A

$$\therefore \text{Rate} = K'[A]$$

$$\text{where } K' = K[H_2O]$$

$$K' = \frac{2.303}{6.93} \log \frac{1}{1/4} = \frac{1}{5}$$

$$\text{Also } [H_2O] = 55.5$$

$$\therefore K = \frac{K'}{[H_2O]} = \frac{1}{5 \times 55.5} = 3.6 \times 10^{-3}$$

Q.10 (B)

$$K = Ae^{-E_a/RT}$$

$$= 1.25 \times 10^8 e^{-\frac{16}{2 \times 10^{-3} \times 400}} = 1.25 \times 10^8 e^{-20}$$

$$= \frac{1.25 \times 10^8}{5 \times 10^8} = \frac{1}{4} \text{ sec}^{-1}$$

$$t_{\text{avg}} = \frac{1}{K} = 4 \text{ sec}$$

Q.11 (A)

Q.12 (C)

$$\text{Rate law} = \frac{dp}{dt} = K_2 [R] [C]$$

$$\text{at steady state } \frac{d[R]}{dt} = 0$$

$$\frac{d[R]}{dt} = K_1 [A] - K_2 [R] [B] - K_2 [R] [C] = 0$$

$$K_1 [A] = \{K_2 [B] + K_2 [C]\} [R]$$

$$[R] = \frac{K_1 [A]}{K_2 \{[B] + [C]\}}$$

$$\text{Rate} = K_2 \times \frac{K_1 [A]}{K_2 [B] + [C]} \times [C]$$

$$= \frac{K_1 [A]}{\frac{[B]}{[C]} + 1}$$

where C is very high

$$\text{Rate} = K_1 [A]$$

$$\therefore \text{order of reaction} = 1$$



Q.13 (C)

$$K = A e^{-\frac{E_a}{RT}}$$

$$\ln K = \ln A - \frac{E_a}{RT}$$

Q.14 (A)

At Steady State

$$K_1[A] = K_2[B] \text{ and } K_3[C] = K_2[B]$$

$$\therefore K_1[A] = K_3[C]$$

$$\frac{[A]}{[C]} = \frac{K_3}{K_1} = 40000$$

Q.15 (B)

Q.16 (A)

Q.17 (C)



$$r = k[\text{CH}_4][\bullet\text{OH}] \quad \text{where } k = 4 \times 10^6 \text{ and } [\bullet\text{OH}] \text{ constant}$$

$\therefore$ Reaction is pseudo 1st order

$$r = k'[\text{CH}_4]$$

$$k' = k[\bullet\text{OH}]$$

$$= 4 \times 10^6 \times 1 \times 10^{-15} = 4 \times 10^{-9}$$

$$\therefore t_{av} = \frac{1}{k'} = \frac{1}{4 \times 10^{-9}} = 25 \times 10^7 \text{ sec.}$$

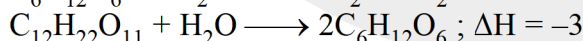
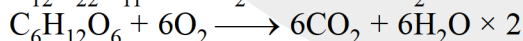
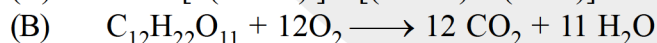
(ii&iii) t_{av} of CH_4 is 25×10^7 sec.

$$\text{and } t_{av} \text{ of isoprene} = \frac{1}{2 \times 10^{11} \times 10^{-15}} = 4000 \text{ sec.}$$

t_{av} of CH_4 is much more than isoprene, there for it will travel longer distance as well as remain for longer time in atmosphere.

Q.18 (A) (B) (D)

(A) $\Delta H = [2(-1263)] - [(-2238) + (-285)] = -3 \text{ kJ}$



Hence, $\Delta_c H_{\text{C}_{12}\text{H}_{22}\text{O}_{11}}$ is greater than $2 \times \Delta_c H_{\text{C}_6\text{H}_{12}\text{O}_6}$

(C) Hydrolysis is exothermic $T \uparrow$ $h \downarrow$

(D) Concentration becomes one-fourth in each equal time interval Hence, 1st order

Q.19 (A) (D)

Q.20 (A) P (B) Q,R (C) S (D) T



Q.21 1202 or 0240

Reaction is of 2nd order because $t_{1/2} \propto \frac{1}{a}$
initial $t_{1/2} = 120$ min.

Q.22 (a) 1, (b) $6.93 \times 10^{-3} \text{ min}^{-1}$, (c) 200, (d) 950 mm

Q.23 $(0.4)^{3/2} = 0.253$

Q.24 0.0805

Q.26 (a) 1, (b) $6.93 \times 10^{-3} \text{ min}^{-1}$, (c) 200, (d) 950 mm

Q.27 0050.00

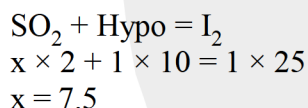
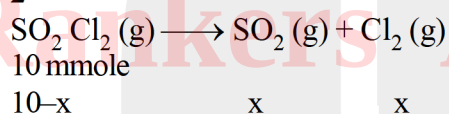
Since B is in large excess, rate law will become
Rate = $k'[A]^2$ where $k' = k[B]$

$$\frac{1}{[A]_t} - \frac{1}{[A]_0} = k't$$

$$\frac{1}{[A]_0 k'} = t_{1/2}$$

$$t_{1/2} = \frac{1}{0.2 \times 5 \times 10^{-5} \times 2 \times 10^3} = 50 \text{ minutes}$$

Q.28 2



Q.29 0006

$$\frac{K_{500}}{K'_{500}} = e^{\frac{693}{1000}}$$

$$K'_{500} = K_{500} \cdot e^{\frac{693}{1000}} = K_{500} \times e^{.693} \quad [\because e^{0.693} = e^{\ln 2}]$$

$$K'_{500} = 2K_{500}$$

$$K_{500} = \frac{\ell n 2}{6}$$

For 75% reaction

$$K'_{500} \times t = \ell n 4$$

$t = 6$ min.

Q.30 4

$$\left[\begin{array}{l} \text{For } n^{\text{th}} \text{ order } (n > 1) \text{ reaction} \\ \frac{1}{[A]^{n-1}} \text{ vs time graph is a straight line always.} \end{array} \right]$$

$\frac{1}{[A]^4}$ vs time graph is given straight line

$$\therefore n - 1 = 4$$

$$n = 5$$

Reaction is of 5th order

$$\therefore -\frac{dA}{dt} = K_A [A]^5$$

$$\frac{dA}{dt} = K_A [A]^5$$

$$\int_{[A]_0}^{[A]} \frac{d[A]}{[A]^5} = -K_A \int_0^t dt$$

$$\Rightarrow -\frac{1}{4} \left[\frac{1}{[A]^4} - \frac{1}{[A]_0^4} \right] = -K_A t$$

$$\frac{1}{[A]^4} = \frac{1}{[A]_0^4} + 4K_A t$$

$$\therefore 4K_A = \tan 45^\circ = 1$$

$$K_A = 1/4 \quad \therefore K_R = \frac{K_A}{2} = \frac{1}{8}$$

$$\therefore \text{ROR} = K_R [A]^5 = \frac{1}{8} [2]^5 = 0004 \text{ Ans.}$$